

DevOps Series

Configuring a PostgreSQL Master-Slave Setup Using Ansible

PostgreSQL is a free and open source, ACID-compliant, transactional database written in the C programming language. It supports updatable views, triggers, stored procedures and foreign keys, and manages concurrency using multi-version concurrency control (MVCC). In this 23rd article in the DevOps series, we will learn how to install and configure a PostgreSQL master-slave replication setup.



PostgreSQL has a number of interfaces for various programming languages and libraries that are available to interact with the database. The primary command-line tool to interact with the database is *psql*. GUI administrative tools are also available. It was first released in 1996 under the PostgreSQL licence.

CentOS 7 (x86_64) is used as the base OS for the PostgreSQL database server. A couple of guest CentOS VMs are launched using KVM. One instance is used as the database master, while the other is used as a replication slave. The *centos* users in both the VMs are given sudo access using the *visudo* command. SELinux is disabled for the exercise.

The host system is a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0 as indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
```

```
configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
ansible python module location = /usr/lib/python3.6/site-
packages/ansible
executable location = /usr/bin/ansible
python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

The Ansible inventory and playbook file are created on the host system as follows:

```
ansible/inventory/kvm/
  /playbooks/configuration/
```

The *inventory/kvm/inventory* file contains the following code:

```
[pgmaster]
host1 ansible_host=192.168.122.174 ansible_connection=ssh
ansible_user=centos ansible_password=centos123
```